

# MULYANKAN

## EVIDENCE-GATED AI PROPOSAL SCRUTINY

Preliminary Science & Technology proposal review support

### WHAT IT DOES

**Turns proposal documents into traceable evidence, rule findings and advisory reviewer reports.**

**Domain: Education & AI**

**Team: Algonauts**

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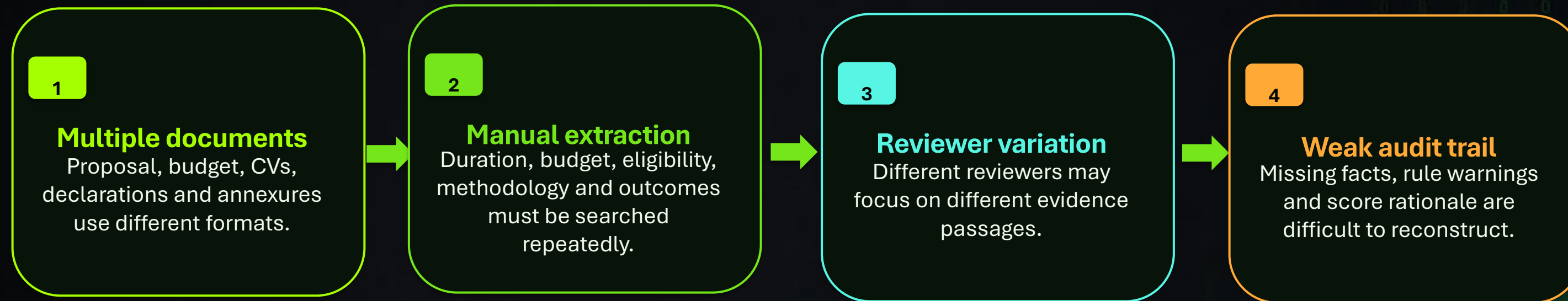


# THE PROBLEM

Preliminary scrutiny is evidence-heavy, document-heavy and hard to standardise.

## Core issue

Experts should judge proposals not spend time hunting for scattered facts.



## Desired institutional capability

Controlled intake → evidence-linked extraction → explicit rule states → reviewer-verifiable advisory report



## WHY THIS PROBLEM EXISTS

The current approach turns proposal review into repeated evidence collection instead of focused judgement.

### CAUSE 1

**Unstructured proposal packages**

### CAUSE 2

**Manual fact extraction**

### CAUSE 3

**No standard evidence contract**

**FRICTION**

**DELAY /  
COST**

**LOW REVIEWER  
CONFIDENCE**

**POOR AUDITABILITY**

### CORE PROBLEM

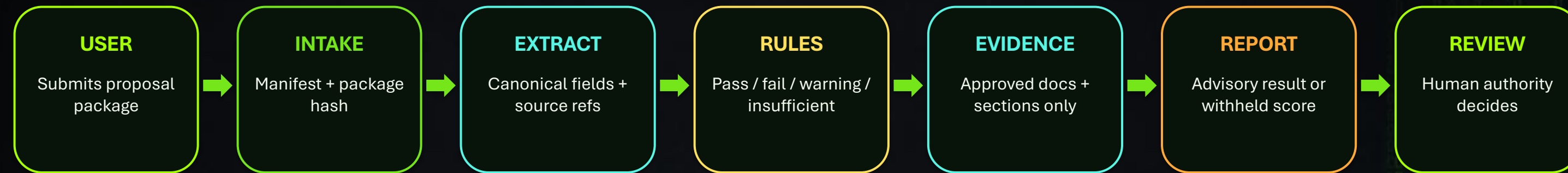
**Important proposal facts are not captured, checked and cited in one controlled workflow.**

**Current method → friction → delay / risk → inconsistent preliminary outcome**

# OUR SOLUTION

Introducing Mulyankan: an evidence-gated, human-in-the-loop scrutiny platform.

**Mulyankan enables reviewers to verify evidence quickly by converting proposal packages into structured fields, rule findings, cited evidence and advisory reports.**



## 01 Controlled package

File roles, hashes and confirmed submission prevent silent document drift.

## 02 Evidence contracts

Each criterion uses only permitted documents and sections.

## 03 Score release gate

Weak evidence leads to withheld/unsupported status, not manufactured confidence.

**Design principle: organise evidence first; score only when evidence coverage allows.**



# PROTOTYPE / PRODUCT EXPERIENCE

A clear reviewer journey from proposal upload to traceable advisory output.

**01 — CREATE** Applicant creates a proposal version and selects the scheme.

**02 — UPLOAD** Files are uploaded into mandatory / optional document slots.

**03 — VERIFY** System checks file type, size, ownership and SHA-256 hash.

**04 — PROCESS** Extractor identifies sections, tables and canonical fields.

**05 — REVIEW** Reviewer receives rule states, evidence citations and advisory findings.

## Large prototype principle

Show real workflow states and report logic; avoid fake screenshots until UI captures are available.

Input → Verified package → Processing → Evidence report → Human review

# WHAT MAKES US DIFFERENT?

Mulyankan is differentiated by governed evidence use and transparent abstention — not by saying “AI-powered”.

| DIMENSION               | CURRENT / BASIC APPROACH                         | MULYANKAN                                                   |
|-------------------------|--------------------------------------------------|-------------------------------------------------------------|
| Evidence source control | Reviewer manually searches across files          | Criterion-level approved docs + sections                    |
| Rule handling           | Eligibility issues can be buried in prose        | Explicit pass / fail / warning / insufficient states        |
| Weak evidence           | Unclear whether missing evidence affected result | Score withheld or marked unsupported with reason            |
| Traceability            | Difficult to reconstruct why a warning appeared  | Package hash, file ID, page ref, rule-set and model version |
| Human authority         | Recommendation may look like a final decision    | Advisory output separated from competent authority          |

Most important line: Mulyankan does not ask only “What does the model think?” — it asks “What evidence was it allowed to use?”



# TECHNICAL ARCHITECTURE

A professional, auditable stack connecting UI, orchestration, evidence storage and advisory intelligence.

## FRONTEND

React interface for submission, tracking, reports and reviewer dashboards

## BACKEND

FastAPI for authentication, submission checks and evaluation orchestration

## PROCESSING

Redis + ARQ worker for background extraction and evaluation jobs

## DATA / STORAGE

PostgreSQL for structured records; MinIO for private proposal documents

## AI / ML

Model registry + artifact hash checks before advisory scoring

**Applicant**

submits

**React UI**

captures

**FastAPI**

orchestrates

**Worker**

extracts

**DB + MinIO**

stores

**Report**

explains

**Trust layer: file hashes, package hash, audit events, rule versions, evidence-contract versions and model identity.**

# FEASIBILITY + IMPACT

Built with available technologies and designed to improve reviewer speed, consistency and auditability.

## FEASIBILITY

- React / FastAPI / PostgreSQL / MinIO / Redis stack is implementable and scalable.
- Existing document parsing handles digital PDF, DOCX and text files; OCR can be added where approved.
- Deterministic rule states are easier to audit than free-form model judgement.
- Model registry and artifact hash checks support version control before scoring.

## IMPACT

- Faster preliminary scrutiny through standardised extraction.
- Better consistency because criteria cite allowed evidence sources.
- Improved audit trail for missing information, rule warnings and advisory rationale.
- Reduced false confidence by withholding scores when evidence is insufficient.

Prototype



Reviewer MVP



Reviewer Pilot



Scale to Schemes



# IMPLEMENTATION ROADMAP

Clear separation between what is demonstrable now and what should be hardened before real deployment.

## NOW

### Hackathon Prototype

- Controlled package upload
- Extraction + rule flow
- Evidence-linked report
- Functional reviewer demo

## NEXT

### Reviewer MVP

- Real UI screenshots
- Scheme configuration
- Reviewer feedback loop
- Reliability improvements

## LATER

### Reviewer Pilot

- Historical/demo proposal cases
- Domain-expert review
- Policy-owner feedback
- Security + data handling

## SCALE

### Institutional Rollout

- More schemes
- More document roles
- Analytics dashboard
- Deployment + integrations

**Critical point: show built workflow honestly; label future hardening clearly.**

# TEAM + EXECUTION

Why this team can execute the solution and what support is needed next.

## WHY WE CAN BUILD THIS

The team combines software engineering, AI/ML thinking and workflow design to convert complex proposal packages into auditable reviewer support.

## NEXT STEP REQUEST

Review the prototype workflow, provide scheme-rule feedback, and help define a controlled reviewer-facing MVP.

### AI-assisted extraction

Canonical fields + uncertainty

### Governed rules

Pass / fail / warning / insufficient

### Evidence-first reports

Cited rationale for reviewers

### Human authority

Decision remains institutional

**Outcome sought: a clear, professional path from working prototype to safe institutional decision-support readiness.**

# THANK YOU

Mulyankan 0.8.0 | Evidence-gated AI proposal scrutiny